

Part 1: Rewrite the given expression as a single power series

whose general term involves x^k .

$$1) \sum_{n=1}^{\infty} 2nC_n x^{n-1} + \sum_{n=0}^{\infty} 6C_n x^{n+1}$$

Let $k=n-1 \therefore n=k+1$
 Let $k=n+1 \therefore n=k-1$

$$= \sum_{k=0}^{\infty} 2(k+1)C_{k+1} x^k + \sum_{k=1}^{\infty} 6C_{k-1} x^k$$

$$= 2C_1 + \sum_{k=1}^{\infty} [2(k+1)C_{k+1} + 6C_{k-1}] x^k \quad \#$$

Part 2: Find two power series solutions of the given differential equation about the ordinary point $x=0$

(Please find and list up the coefficients up to the x^7 -term)

$$2) y'' - 3xy = 0$$

$\Rightarrow x=0$ is ordinary point

$\therefore$ 存在 Taylor 級數解

$$\Rightarrow y = \sum_{n=0}^{\infty} a_n x^n, \quad |x| < L = \infty$$

$$\Rightarrow y' = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

$$\Rightarrow y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

$$\Rightarrow y'' - 3xy = 0$$

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} - 3 \sum_{n=0}^{\infty} a_n x^{n+1} = 0$$

$$\sum_{k=0}^{\infty} (k+2)(k+1) a_{k+2} x^k - 3 \sum_{k=1}^{\infty} a_{k-1} x^k = 0$$

$$2a_2 + \sum_{k=1}^{\infty} [(k+2)(k+1) a_{k+2} - 3a_{k-1}] x^k = 0$$

$$\text{For } k=0, \quad 2a_2 = 0 \quad \therefore a_2 = 0$$

$$\text{For } k \geq 1, \quad (k+2)(k+1) a_{k+2} - 3a_{k-1} = 0$$

$$a_{k+2} = \frac{3a_{k-1}}{(k+2)(k+1)}$$

$$\Rightarrow a_0 = a_0 (\text{arbitrary}), \quad a_1 = a_1 (\text{arbitrary}), \quad a_2 = 0$$

$$\Rightarrow k=1: a_3 = \frac{3}{3 \times 2} a_0 = \frac{1}{2} a_0$$

$$k=2: a_4 = \frac{3}{4 \times 3} a_1 = \frac{1}{4} a_1$$

$$k=3: a_5 = \frac{3}{5 \times 4} a_2 = 0$$

$$k=4: a_6 = \frac{3}{6 \times 5} a_3 = \frac{1}{10} a_3 = \frac{1}{10} \left(\frac{1}{2} a_0 \right) = \frac{1}{20} a_0$$

$$k=5: a_7 = \frac{3}{7 \times 6} a_4 = \frac{1}{14} a_4 = \frac{1}{14} \left(\frac{1}{4} a_1 \right) = \frac{1}{56} a_1$$

$$\therefore y = a_0 y_1 + a_1 y_2$$

$$y_1 = 1 + \frac{x^3}{2} + \frac{x^6}{20} + \dots$$

$$y_2 = x + \frac{x^4}{4} + \frac{x^7}{56} + \dots \quad \#$$

Part 3: Use the power series method to solve the given initial-value problem

$$3) (x-1)y'' - xy' + y = 0, \quad y(0) = -2, \quad y'(0) = 6$$

$\Rightarrow \therefore x=0$ is an ordinary point

$\therefore$ There exists a Taylor series solution

$$\Rightarrow y(x) = \sum_{n=0}^{\infty} a_n x^n$$

$$y'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

$$y''(x) = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

$$\Rightarrow (x-1) \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} - x \sum_{n=1}^{\infty} n a_n x^{n-1} + \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-1} - \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} - \sum_{n=1}^{\infty} n a_n x^n + \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\sum_{n=1}^{\infty} n(n+1) a_{n+1} x^n - \sum_{n=0}^{\infty} (n+1)(n+2) a_{n+2} x^n - \sum_{n=1}^{\infty} n a_n x^n + \sum_{n=0}^{\infty} a_n x^n = 0$$

$$-2a_2 + a_0 + \sum_{n=1}^{\infty} [n(n+1) a_{n+1} - (n+1)(n+2) a_{n+2} - (n-1) a_n] x^n = 0$$

$$\therefore a_{n+2} = \frac{n(n+1) a_{n+1} - (n-1) a_n}{(n+1)(n+2)}$$

$$\Rightarrow \text{From } y(x), \text{ subs } x=0 \quad \therefore a_0 = -2$$

$$\Rightarrow \text{From } y'(x), \text{ subs } x=0 \quad \therefore a_1 = 6$$

$$\Rightarrow \text{For } n=0: a_2 = \frac{1}{2} a_0$$

$$n=1: a_3 = \frac{2}{2 \times 3} a_2 = \frac{1}{3} a_2 = \frac{1}{3 \times 2} a_0$$

$$n=2: a_4 = \frac{6a_3 - a_2}{3 \times 4} = \frac{1}{4 \times 3 \times 2} a_0$$

$$n=3: a_5 = \frac{12a_4 - 2a_3}{4 \times 5} = \frac{1}{5 \times 4 \times 3 \times 2} a_0$$

$$\Rightarrow y(x) = a_0 + a_1 x + \frac{1}{2} a_0 x^2 + \frac{1}{6} a_0 x^3 + \frac{1}{24} a_0 x^4 + \frac{1}{120} a_0 x^5 + \dots$$

$$= 6x - 2 \left(1 + \frac{1}{2!} x^2 + \frac{1}{3!} x^3 + \frac{1}{4!} x^4 + \frac{1}{5!} x^5 + \dots + \frac{1}{n!} x^n \right)$$

$$= 6x - 2 \left(1 + x + \frac{1}{2!} x^2 + \frac{1}{3!} x^3 + \frac{1}{4!} x^4 + \frac{1}{5!} x^5 + \dots + \frac{1}{n!} x^n \right)$$

$$= 8x - 2e^x \quad \#$$